

Self-Attention Numerical Example

1 Setup and Initial Dimensions

We assume a sequence length of 3 words, an initial static embedding size (d_{model}) of 4, and an attention dimension (d_k) of 3.

1.1 Input Embeddings (X)

The 3 words are represented as rows in matrix X :

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Where:

- Word 1 (x_1) = [1, 0, 1, 0]
- Word 2 (x_2) = [0, 2, 0, 1]
- Word 3 (x_3) = [1, 1, 1, 1]

1.2 Weight Matrices (W_Q, W_K, W_V)

The learned projection weights are defined as:

$$W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad W_K = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

2 Generating Queries, Keys, and Values

Multiplying the input matrix X by each weight matrix yields our attention vectors.

2.1 Queries ($Q = X \times W_Q$)

$$Q = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 1 \\ 2 & 2 & 2 \end{bmatrix}$$

- q_1 = [2, 0, 1]
- q_2 = [0, 3, 1]
- q_3 = [2, 2, 2]

2.2 Keys ($K = X \times W_K$)

$$K = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 0 & 2 \\ 2 & 3 & 2 \end{bmatrix}$$

- $k_1 = [0, 2, 1]$
- $k_2 = [3, 0, 2]$
- $k_3 = [2, 3, 2]$

2.3 Values ($V = X \times W_V$)

$$V = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 3 & 3 \\ 2 & 4 & 2 \end{bmatrix}$$

- $v_1 = [2, 2, 0]$
- $v_2 = [0, 3, 3]$
- $v_3 = [2, 4, 2]$

3 Calculating Attention for Word 1

3.1 Step 1: Raw Dot Products (Scores)

We dot product the query of Word 1 (q_1) with all key vectors:

- $\text{Score}_{1,1} = q_1 \cdot k_1 = (2 \times 0) + (0 \times 2) + (1 \times 1) = 1$
- $\text{Score}_{1,2} = q_1 \cdot k_2 = (2 \times 3) + (0 \times 0) + (1 \times 2) = 8$
- $\text{Score}_{1,3} = q_1 \cdot k_3 = (2 \times 2) + (0 \times 3) + (1 \times 2) = 6$

3.2 Step 2: Scaling the Scores

We scale by dividing by $\sqrt{d_k} = \sqrt{3} \approx 1.732$:

- Scaled $\text{Score}_1 = 1/1.732 \approx 0.577$
- Scaled $\text{Score}_2 = 8/1.732 \approx 4.619$
- Scaled $\text{Score}_3 = 6/1.732 \approx 3.464$

3.3 Step 3: Softmax Transformation

We convert the scaled scores into probabilities:

$$\begin{aligned} e^{0.577} &\approx 1.781 \\ e^{4.619} &\approx 101.393 \\ e^{3.464} &\approx 31.945 \\ \text{Sum} &= 1.781 + 101.393 + 31.945 = 135.119 \end{aligned}$$

The final Softmax weights are:

- $\text{Weight}_1 = 1.781/135.119 \approx \mathbf{0.013}$
- $\text{Weight}_2 = 101.393/135.119 \approx \mathbf{0.750}$
- $\text{Weight}_3 = 31.945/135.119 \approx \mathbf{0.236}$

4 Final Contextual Output for Word 1

The final output is the weighted sum of the value vectors:

$$\text{Output}_1 = (\text{Weight}_1 \times v_1) + (\text{Weight}_2 \times v_2) + (\text{Weight}_3 \times v_3)$$

Vector	Weight	Value Vector	Weighted Result
v_1	0.013	[2, 2, 0]	[0.026, 0.026, 0.000]
v_2	0.750	[0, 3, 3]	[0.000, 2.250, 2.250]
v_3	0.236	[2, 4, 2]	[0.472, 0.944, 0.472]

Summing the rows yields the final contextual vector for Word 1:

$$\text{Output}_1 = [0.498, 3.220, 2.722]$$